Notes and Solutions

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0.1 GFKG: Maxwell's Equations

1. *Proof (electromagnetic plane wave).* Let $\vec{k}$ be a vector in $\mathbb{R}^3$ and let $\omega = |\vec{k}|$. Fix $\vec{E} \in \mathbb{C}^3$ with $\vec{k} \cdot \vec{E} = 0$ and $i\vec{k} \times \vec{E} = \omega \vec{E}$. We will show that

$$\vec{\mathcal{E}}(t, \vec{x}) = \vec{E} e^{-i(\omega t - \vec{k} \cdot \vec{x})}, \qquad \text{for } \vec{k} = \begin{pmatrix} k^x \\ k^y \\ k^z \end{pmatrix} \text{ and } \vec{E} = \begin{pmatrix} E^x \\ E^y \\ E^z \end{pmatrix},$$

satisfies the vacuum Maxwell equations: $\nabla \cdot \vec{\mathcal{E}} = 0$ and $\nabla \times \vec{\mathcal{E}} = i \dfrac{\partial \vec{\mathcal{E}}}{\partial t}$.

We compute the partial derivative with respect to time below:

$$\frac{\partial \vec{\mathcal{E}}}{\partial t} = \vec{E} e^{-i(-\vec{k} \cdot \vec{x})} \frac{\partial}{\partial t} \left(e^{-i(\omega t)} \right) = \vec{E} e^{-i(-\vec{k} \cdot \vec{x})} \left(-i\omega e^{(-i\omega)t} \right) = -i\omega \vec{\mathcal{E}}.$$

and of a spatial component $\mu = x, y,$ or z with respect to a spatial component z below:

$$\begin{aligned} \frac{\partial \mathcal{E}^\mu}{\partial z} &= E^\mu e^{-i(\omega t - k^x x - k^y y)} \frac{\partial}{\partial z} \left(e^{-i(-k^z z)} \right) = E^\mu e^{-i(\omega t - k^x x - k^y y)} \left(i k^z e^{(ik^z) z} \right) \\ &= i k^z E^\mu e^{-i(\omega t - \vec{k} \cdot \vec{x})}. \end{aligned}$$

The divergence is then given by:

$$\begin{aligned} \nabla \cdot \vec{\mathcal{E}} &= \frac{\partial \mathcal{E}^x}{\partial x} + \frac{\partial \mathcal{E}^y}{\partial y} + \frac{\partial \mathcal{E}^z}{\partial z} = i k^x E^x e^{-i(\omega t - \vec{k} \cdot \vec{x})} + i k^y E^y e^{-i(\omega t - \vec{k} \cdot \vec{x})} + i k^z E^z e^{-i(\omega t - \vec{k} \cdot \vec{x})} \\ &= i \left(k^x E^x + k^y E^y + k^z E^z \right) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = i \left(\vec{k} \cdot \vec{E} \right) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = i \left(0 \right) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = 0, \end{aligned}$$

and the curl is given by:

$$\begin{aligned} \nabla \times \vec{\mathcal{E}} &= \begin{pmatrix} \frac{\partial \mathcal{E}^z}{\partial y} - \frac{\partial \mathcal{E}^y}{\partial z} \\ \frac{\partial \mathcal{E}^x}{\partial z} - \frac{\partial \mathcal{E}^z}{\partial x} \\ \frac{\partial \mathcal{E}^y}{\partial x} - \frac{\partial \mathcal{E}^x}{\partial y} \end{pmatrix} = i \begin{pmatrix} k^y E^z - k^z E^y \\ k^z E^x - k^x E^z \\ k^x E^y - k^y E^x \end{pmatrix} e^{-i(\omega t - \vec{k} \cdot \vec{x})} \\ &= i(\vec{k} \times \vec{E}) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = (\omega \vec{E}) e^{-i(\omega t - \vec{k} \cdot \vec{x})} = i \left(-i\omega \vec{E} e^{-i(\omega t - \vec{k} \cdot \vec{x})} \right) = i \frac{\partial \vec{\mathcal{E}}}{\partial t}. \end{aligned}$$

Therefore, the plane wave $\vec{\mathcal{E}}$ satisfies the vacuum Maxwell equations. □

0.2 Manifolds

2. *Proof (continuity).* Consider a function $f : \mathbb{R}^n \to \mathbb{R}^m$ We show that f is continuous according to the open pre-image definition if and only if it is according to the epsilon-delta definition.

   First, recall that open n-dimensional and m-dimensional balls are basis elements of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively.

Suppose that f satisfies the open pre-image definition: given any open set $U \subseteq \mathbb{R}^m$, the inverse image $f^{-1}U \subseteq \mathbb{R}^n$ is open. Consider some $x \in \mathbb{R}^n$ and $\varepsilon > 0$. Then, there are open sets

$$V = B_{\mathbb{R}^m}(f(x), \varepsilon) = \{z \in \mathbb{R}^m \mid ||z - f(x)|| < \varepsilon\} \subseteq \mathbb{R}^m \quad \text{and} \quad f^{-1}V = \bigcup_{y \in f^{-1}B} B(y, \delta_y) \subseteq \mathbb{R}^n$$

with $f(x) \in V$. As a result, there exists $\delta = \delta_x > 0$ such that $x \in B(x, \delta) \subset f^{-1}V$. Thus, $y \in B(x, \delta)$, i.e. $||y - x|| < \delta$, implies that $f(y) \in V$ and thus $||f(y) - f(x)|| < \varepsilon$.

Now suppose instead that f satisfies the epsilon-delta definition: for all $x \in \mathbb{R}^m$ and all $\varepsilon > 0$, there exists $\delta > 0$ such that $||y - x|| < \delta$ implies $||f(y) - f(x)|| < \varepsilon$. Consider an open set $W \subset \mathbb{R}^m$, and let $W = U \cup V$ for $U \cap f(\mathbb{R}^n) = \emptyset$ and $V \subset f(\mathbb{R}^n)$. Then,

$$V = \{z \in W \mid z \in f(\mathbb{R}^n)\} = \bigcup_{f(x) \in V} B(f(x), \varepsilon_x) \quad \text{and} \quad f^{-1}V = \bigcup_{f(x) \in V} f^{-1}(B(f(x), \varepsilon_x))$$

If $x \in f^{-1}V$, then there exists $\delta > 0$ such that $f(x) \in f(B(x, \delta)) \subseteq B(f(x), \varepsilon)) \subseteq V$, so $x \in B(x, \delta)$ and $B(x, \delta) \subseteq f^{-1}V$ yield

$$f^{-1}V = \bigcup_{x \in f^{-1}V} B(x, \delta_x).$$

Thus, $f^{-1}W = f^{-1}U \cup f^{-1}V$ is the union of an open set $f^{-1}V$ and an open set $f^{-1}U = \emptyset$.

□

3. *Proof (sphere manifold).* Let S^n, the n-sphere, be the unit sphere in $\mathbb{R}^{n+1}$

$$S^n = \{\vec{x} \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} (x^i)^2 = 1\}$$

with its induced topology.

It is easy to prove that S^n is an $(n+1)$-dimensional manifold, so we will show it is in fact an n-dimensional manifold by covering it with two open sets $N, S \subset S^n$

$$N = U_N = S^n \setminus \{n\} \qquad \text{and} \qquad S = U_S = S^n \setminus \{s\},$$

where $n = (0, \ldots, +1)$ and $s = (0, \ldots, -1)$ are the north and south poles, respectively. Since $\{n\}, \{s\}$ are singletons, they are closed in $\mathbb{R}^{n+1}$ and in S^n, so their complements N and S in S^n are thus open. We define an atlas $\{\varphi_N : N \to \mathbb{R}^n, \varphi_S : S \to \mathbb{R}^n\}$ via the stereographic projections component-wise for $1 \leq i \leq n$

$$\varphi_N^i(x) = \frac{x_i}{1 - x_n}, ({\varphi_S}^{-1}(y))^i = \frac{2y_i}{+1 + \sum_{i=1}^n y_i^2} \quad \text{and} \quad \varphi_S^i(x) = \frac{x_i}{1 + x_n}, ({\varphi_S}^{-1}(y))^i = \frac{2y_i}{-1 + \sum_{i=1}^n y_i^2},$$

and define the $(n+1)^{\text{th}}$ component of the inverse as follows:

$$({\varphi_N}^{-1}(y))^{n+1} = \frac{-1 + \sum_{i=1}^n y_i^2}{+1 + \sum_{i=1}^n y_i^2}.$$

Notice that the transition function $\varphi_S \circ {\varphi_N}^{-1}$ is smooth and well-defined because it is a composition of smooth operations; the denominator of ${\varphi_n}^{-1}$ is the positive sum of $+1$ and a positive sum of squares, and the denominator of φ_S is only zero on s, on which it is not defined. Similarly, the transition function $\varphi_S \circ {\varphi_N}^{-1}$ is smooth where it is defined.

Therefore, S^n with its induced topology is a manifold. □

4. *Proof (induced manifold).* We show that if M is a manifold with charts $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ and U is an open subset of M, then U with its induced topology is a manifold.

   We equip U with charts $\psi_\alpha : V_\alpha \to \mathbb{R}^n$, where $V_\alpha = U_\alpha \cap U$ are open sets covering $M \cap U = U$ and where $\psi_\alpha = \varphi_\alpha|_{V_\alpha}$. Then, the transition functions $\psi_\alpha \circ {\psi_\beta}^{-1}$ inherit smoothness via restriction wherever defined. □

5. *Proof (product manifold).* We show that if M and N are m- and n-dimensional manifolds with charts $\varphi_\alpha : U_\alpha \to \mathbb{R}^m$ and $\psi_\beta : V_\beta \to \mathbb{R}^n$ for $\alpha \in A$ and $\beta \in B$, then $M \times N$ is an $(m+n)$-dimensional manifold.

   We can cover the product topology $M \times N$ with open sets $W_\gamma = U_\alpha \times V_\beta$ for $\gamma = \gamma_{\alpha,\beta} \in C = A \times B$, so we define charts $\varphi_\alpha \times \psi_\beta = \xi_\gamma : W_\gamma \to \mathbb{R}^{m+n}$ component-wise. Then, the transition functions $\xi_\gamma \circ {\xi_\delta}^{-1}$ inherit smoothness component-wise wherever defined. □

6. *Proof.* We show that if M and N are n-dimensional manifolds with charts $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ and $\psi_\beta : V_\beta \to \mathbb{R}^n$ for $\alpha \in A$ and $\beta \in B$, then $M \cup N$ is an n-dimensional manifold.

   We can cover the disjoint union topology $M \cup N$ with open sets $W_\gamma = U_\alpha \cup V_\beta$ for $\gamma = \gamma_{\alpha,\beta} \in C = A \times B$, so we define charts $\xi_\gamma : W_\gamma \to \mathbb{R}^n$ piece-wise by $\xi_\gamma|_{U_\alpha} = \varphi_\alpha$ and $\xi_\gamma|_{V_\beta} = \psi_\beta$. Then, transition functions $\xi_\gamma \circ {\xi_\delta}^{-1}$ inherit smoothness piece-wise wherever defined. □

0.3 Vector Fields

7. *Proof (addition and function multiplication).* We will show that $v+w$ and gw are in $\mathrm{Vect}(M)$ for all $v, w \in \mathrm{Vect}(M)$ and $g \in C^\infty(M)$.

   Let $f, g, h \in C^\infty(M)$, and $\alpha \in \mathbb{R}^n$. Then,

$$
\begin{aligned}
(v+w)(fg) &= v(fg) + w(fg) = v(f)g + fv(g) + w(f)g + fw(g) = (v+w)(f)g + f(v+w)(g),\\
(v+w)(f+g) &= v(f+g) + w(f+g) = v(f) + v(g) + w(f) + w(g) = (v+w)(f) + (v+w)(g)\\
(v+w)(\alpha f) &= v(\alpha f) + w(\alpha f) = \alpha v(f) + \alpha w(f) = \alpha\left(v(f) + w(f)\right) = \alpha(v+w)(f), \text{ and}\\
(gw)(f+h) &= gw(f+h) = gw(f) + gw(h) = (gw)(f) + (gw)(h)\\
(gw)(fh) &= g\left(w(f)h + fw(h)\right) = (gw)(f)h + f(gw)(h)\\
(gw)(\alpha f) &= gw(\alpha f) = \alpha g w(f) = \alpha(gw)(f).
\end{aligned}
$$

   Therefore, $v+w$ and gw are indeed vector fields on M. □

8. *Proof (module of vector fields).* Let $v, w \in \mathrm{Vect}(M)$ and $f, g, h \in C^\infty(M)$. Then, we obtain

$$
\begin{aligned}
(f(v+w))(h) &= f(v+w)(h) = f(v(h) + w(h)) = fv(h) + fw(h)\\
((f+g)v)(h) &= (f+g)v(h) = fv(h) + gv(h)\\
((fg)v)(h) &= (fg)v(h) = f(gv)(h), \text{ and}\\
(1v)(h) &= 1v(h) = v(h).
\end{aligned}
$$

   Therefore, $\mathrm{Vect}(M)$ is a module over $C^\infty(M)$. □

9. *Proof (linear independence).* We show that if $v^\mu \partial_\mu = 0$, we must have $v^\mu = 0$ for all μ.

   Suppose $v^\mu \partial_\mu f = 0$ for all $f \in C^\infty(\mathbb{R}^n)$. Then, $v^\mu = 0$ or $\frac{\partial f}{\partial x^\mu} = 0$ for each $\mu = 1, \ldots, n$. Since the result must hold for all f, which in particular may not satisfy the latter condition, it follows that we must have $v^\mu = 0$ for all μ.

   Therefore, the vector fields $\{\partial_\mu\}$ on $\mathbb{R}^n$ are linearly independent. □

10. *Proof (tangent vector equivalence of vector fields).* We will show that $v = w$ if and only if $v_p = w_p$ for all $p \in M$. If $v = w$, then we obtain

$$v_p(f) = v(f)(p) = w(f)(p) = w_p(f) \qquad \text{for all } f \in C^\infty(M), \tag{1}$$

for all $p \in M$. If $v_p = w_p$ for all $p \in M$, then we obtain $v = w$ from the following implication:

$$v(f)(p) = v_p(f) = w_p(f) = w(f)(p) \implies v(f) = w(f) \qquad \text{for all } f \in C^\infty(M). \tag{2}$$

Therefore, every tangent vector at p is of the form v_p for some vector field v. □

11. *Proof (tangent vector space).* We show that T_pM is a vector space over the real numbers.

Consider scalars $a, b \in \mathbb{R}$ and tangent vectors $u_p, v_p, w_p \in T_pM$ at $p \in M$.

Then, we show they satisfy the vector space axioms:

commutativity	$u_p(f) + v_p(f) = u(f)(p) + v(f)(p) = v(f)(p) + u(f)(p) = v_p(f) + u_p(f)$
additive identity	$v_p(f) + 0 = v(f)(p) + 0 = v(f)(p) = v_p(f)$
inverse	$v_p(f) + (-v_p(f)) = v(f)(p) - v(f)(p) = 0$
compatibility	$a(bv_p(f)) = a(bv(f)(p)) = (ab)v(f)(p) = (ab)v_p(f)$
scalar identity	$1v_p(f) = 1v(f)(p) = v(f)(p) = v_p(f)$
vector distributivity	$a(u_p(f) + v_p(f)) = a(u(f)(p) + v(f)(p)) = au(f)(p) + av(f)(p) = au_p(f) + av_p(f)$
field distributivity	$(a + b)v_p(f) = (a + b)v(f)(p) = av(f)(p) + bv(f)(p) = av_p(f) + bv_p(f),$

and associativity: $u_p(f) + (v_p(f) + w_p(f)) = u(f)(p) + (v(f)(p) + w(f)(p)) = (u(f)(p) + v(f)(p)) + w(f)(p) = (u_p(f) + v_p(f)) + w_p(f)$, for all $f \in C^\infty(M)$. □

12. *Proof (tangent curve).* Consider $\gamma : \mathbb{R} \to M$ and define $\gamma'(t) \in \mathrm{Vect}(M)$ as follows:

$$\gamma' : f \mapsto \frac{d}{dt} f(\gamma(t)).$$

For some $t \in \mathbb{R}$, let $p = \gamma(t)$ and $\gamma_p = \gamma'(t)$. Then, linearity and the product rule of differentiation yield:

$$\begin{aligned}
\gamma_p(f + g) &= \frac{d}{dt}(f + g)(\gamma(t)) = \frac{d}{dt} f(\gamma(t)) + \frac{d}{dt} g(\gamma(t)) = \gamma_p(f) + \gamma_p(g) \\
\gamma_p(\alpha f) &= \frac{d}{dt}(\alpha f)(\gamma(t)) = \alpha \frac{d}{dt} f(\gamma(t)) = \alpha \gamma_p(f) \\
\gamma_p(fg) &= \frac{d}{dt}(fg)(\gamma(t)) = \frac{d}{dt}\left(f(\gamma(t))\right) g(\gamma(t)) + f(\gamma(t)) \frac{d}{dt} g(\gamma(t)) = \gamma_p(f) g(p) + f(p) \gamma_p(g).
\end{aligned}$$

Therefore, $\gamma'(t)$ defines a tangent vector at $\gamma(t)$ for each t. □

13. Let $\phi : \mathbb{R} \to \mathbb{R}$ be given by $\phi(t) = e^t$. Let x be the usual coordinate function on $\mathbb{R}$.

Proof. Then, the pullback is given below:

$$\phi^\star x = x(\phi(t)) = x(e^t) = e^x. \tag{3}$$

□

14. Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be rotations counterclockwise by an angle ϕ. Let x, y be the usual coordinate functions on $\mathbb{R}^2$.

Proof. Then, the pullback is given below:

$$\phi^\star x = x(\phi(x,y)) = x\begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix} = (\cos\theta)x - (\sin\theta)y$$
$$\phi^\star y = y(\phi(x,y)) = y\begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix} = (\sin\theta)x + (\cos\theta)y.$$

□

15. *Proof (smooth map).* Let $f \in C^\infty(N)$, $\gamma : \mathbb{R} \to M$, and $\phi : M \to N$ is smooth.

    If ϕ is smooth, then $\phi \circ \gamma : \mathbb{R} \to N$ is smooth iff $f(\phi \circ \gamma) = \phi^\star f(\gamma)$ depends smoothly on t iff γ is smooth. If ϕ is not smooth, then neither implication holds in general. □

16. *Proof (pushforward of tangent curve).* For any $f \in \mathbb{C}^\infty(M)$, we obtain the following:

$$((\phi \circ \gamma)'(t))(f) = \frac{d}{dt} f(\phi \circ g) = \frac{d}{dt} f \circ \phi \circ g = \frac{d}{dt}\phi^\star f(g) = g'(\phi^\star f) = (\phi_\star g')(f),$$

    where $g = \gamma(t)$ and $g' = \gamma'(t)$ for any $t \in \mathbb{R}$. Therefore, $(\phi \circ \gamma)'(t) = \phi_\star(\gamma'(t))$. □

17. *Proof (pushforward linearity).* Consider the pushforward operation $\phi_\star : T_pM \to T_{\phi(p)}N$ for some $p \in M$ and $\phi : M \to N$. For any $v, w \in T_pM$, $\alpha \in R$ and $f \in C^\infty(N)$, the linearity of T_pM yields the following:

$$\begin{aligned} (\phi_\star(v+w))(f) = (v+w)(\phi^\star f) = v(\phi^\star f) + w(\phi^\star f) &= (\phi_\star v)(f) + (\phi_\star w)(f), \\ (\phi_\star(\alpha v))(f) = (\alpha v)(\phi^\star f) = \alpha v(\phi^\star f) &= \alpha(\phi_\star v)f, \end{aligned}$$

    i.e. that $\phi_\star$ is linear. □

18. Let $\phi : M \to N$ and $v \in \mathrm{Vect}(M)$.

    Proof (pushforward vector field). We compute the pushforward at $f \in C^\infty(N)$ below:

$$\begin{aligned} (\phi_\star v)_q(f) &= (\phi_\star v)(f)(q) && \text{by definition of a tangent vector at } q \\ &= v(\phi^\star f)(p) && \text{by definition of pushforward whenever } \phi(p) = q \\ &= v_p(\phi^\star f) = (\phi_\star v_p)(f) && \text{as above.} \end{aligned}$$

    Therefore, $(\phi_\star v)_q = \phi_\star(v_p)$. □

19. Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation counterclockwise by an angle θ. Let ∂_x, ∂_y be the coordinate vector fields on $\mathbb{R}^2$.

    Proof. Let $(x,y) = \phi(\chi, \upsilon) = (\cos(\theta)\chi - \sin(\theta)\upsilon,\ \sin(\theta)\chi + \cos(\theta)\upsilon)$. We evaluate the coordinate vector field pushforwards at $f : \mathbb{R}^2 \to \mathbb{R}$ via the chain rule below.

$$\begin{aligned} (\phi_\star\partial_x)(f) = \partial_x(\phi^\star f) = \partial_x\left(f \circ \phi\begin{pmatrix}\chi \\ \upsilon\end{pmatrix}\right) = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \chi} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \chi} = \cos(\theta)\frac{\partial f}{\partial x} + \sin(\theta)\frac{\partial f}{\partial y} \\ (\phi_\star\partial_y)(f) = \partial_y(\phi^\star f) = \partial_y f\begin{pmatrix}x \\ y\end{pmatrix} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \upsilon} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \upsilon} = -\sin(\theta)\frac{\partial f}{\partial x} + \cos(\theta)\frac{\partial f}{\partial y} \end{aligned}$$

    □

20. Let v be the vector field $x^2\partial_x + y\partial_y$ on $\mathbb{R}^2$.

Proof. Let $\gamma(t) = (x(t), y(t))$. If $\gamma'(t)(f) = \frac{d}{dt} f(\gamma(t)) = v(f)(\gamma(t)) = v_{\gamma(t)}(f)$ for all $f \in C^\infty(M)$, then we obtain:

$$\gamma'(t)(f) = \left(\frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t}\right)(\gamma(t)) = \left(x^2\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y}\right)(\gamma(t)) = v_{\gamma(t)}(f).$$

From $\dot{x}(t) = x(t)^2$ and $\dot{y}(t) = y(t)$, we obtain $x(t) = \frac{1}{c_1 - t}$ and $y(t) = c_2 e^t$ as general solutions with $c_1 = \frac{1}{x(0)} + 0$ and $c_2 = \frac{y(0)}{e^0} = y(0)$, which yields the following particular solution:

$$\gamma(t) = \begin{pmatrix} (1/x(0) - t)^{-1} \\ y(0)e^t \end{pmatrix},$$

which is defined whenever $\frac{1}{x(0)} - t \neq 0$, which holds for all t iff $x(0) = 0$.

Thus, the integral curves $\gamma(t)$ defined for all t are those which cross the origin. □

21. Let $s, r \in \mathbb{R}$.

Proof (flow semigroup). Let $p \in M$ and $\gamma_p : \mathbb{R} \to M$ be the integral curve through $\gamma_p(0) = p$ defined by $\gamma_p : s \mapsto \phi_s(p)$. Then, ϕ_0 is the identity because $\phi_0(p) = \gamma_p(0) = p$. Furthermore, we obtain $\phi_t \circ \phi_s = \phi_{t+s}$ as follows: $\phi_{t+s}(p) = \gamma_p(t+s) = \gamma_{\gamma_s(p)}(t) = \phi_t(\phi_s(p))$. □

22. Consider the normalized vector fields in the r and θ directions on the plane in polar coordinates (not defined at the origin):

$$v = \frac{x\partial_x + y\partial_y}{\sqrt{x^2+y^2}} \tag{4}$$

$$w = \frac{x\partial_x - y\partial_y}{\sqrt{x^2+y^2}} \tag{5}$$

Proof. Let $r = \sqrt{x^2+y^2}$. Then, the commutativity of the ∂_μ yield the following commutator:

$$\begin{aligned}
[v, w] &= \left(\frac{x\partial_x + y\partial_y}{r}\right)\left(\frac{x\partial_x - y\partial_y}{r}\right) - \left(\frac{x\partial_x - y\partial_y}{r}\right)\left(\frac{x\partial_x + y\partial_y}{r}\right) \\
&= \frac{1}{r^2}\left(x\partial_x(x\partial_x - y\partial_y) + y\partial_y(x\partial_x - y\partial_y) - x\partial_x(x\partial_x + y\partial_y) + y\partial_y(x\partial_x + y\partial_y)\right) \\
&= \frac{0+0+0+0}{r^2} = 0.
\end{aligned}$$

□

23. If $v, w \in \mathrm{Vect}M$ generate the flows $\phi_t, \psi_s : M \to M$, then $f \in C^\infty(M)$ gives the following.

$$(vf)(p) = \frac{d}{dt} f(\phi_t(p))\Big|_{t=0} \qquad \text{and} \qquad (wf)(p) = \frac{d}{ds} f(\psi_s(p))\Big|_{s=0}$$

Proof (commutator flow). Since $\phi(p) : \mathbb{R} \to M$ defines a curve $t \mapsto \phi_t(p) = \gamma_p(t)$, we use the tangent vector $\phi_t'(p)(f) = \left(\frac{d}{dt}\phi_t(p)\right)(f) = v_{\phi_t(p)}(f) = v_{\gamma(t)}(f) = \gamma'(t)(f)$ to compute the composite flows below:

$$\begin{aligned}
v(w(f)) &= \frac{d}{dt} wf(\phi_t)\Big|_{t=0} = \frac{d}{dt}\phi_t(wf)\Big|_{t=0} = \frac{d}{dt}\phi_t\left(\frac{d}{ds} f(\psi_s)\Big|_{s=0}\right)\Big|_{t=0} \\
&= \frac{d}{dt}\phi_t\left(\frac{d}{ds}\psi_s(f)\Big|_{s=0}\right)\Big|_{t=0} = \frac{d}{dt}\frac{d}{ds}\psi_s(f(\phi_t))\Big|_{s=0,t=0} = \frac{d^2}{dtds} f(\phi_t(\psi_s))\Big|_{s=t=0},
\end{aligned}$$

and $w(v(f)) = \frac{d^2}{dtds} f(\psi_s(\phi_t))\Big|_{s=t=0}$ follows from symmetry.

We apply subtraction and evaluation to check the following equation:

$$[v,w](f)(p) = (vwf)(p) - (wvf)(p) = \frac{d^2}{dtds} f(\phi_t(\psi_s(p))) - f(\psi_s(\phi_t(p)))\Big|_{s=t=0}. \tag{6}$$

□

24. Let $u, v, w \in \mathrm{Vect}(M)$ and $\alpha, \beta \in \mathbb{R}$.

Proof (commutator identities). Let v, w be as above and let $(uf)(p) = \frac{d}{dt} f(\chi_r(p)) \mid_{r=0}$. Then, we obtain anti-commutativity and linearity as follows:

$$\begin{aligned}
-[w,v](f) &= -\frac{d^2}{dtds}\left[f(\psi_s(\phi_t)) - f(\phi_t(\psi_s))\right]_{s=t=0} = [v,w] \\
[u,\alpha v](f) &= \alpha\frac{d^2}{drdt} f(\chi_r(\phi_t(p))) - \alpha\frac{d^2}{drdt} f(\phi_t(\chi_r(p))) = \alpha[u,v](f) \\
[u,v+w](f) &= \frac{d^2}{drdt} f(\chi_r((\phi_t+\psi_t)(p))) - \frac{d^2}{drdt} f((\phi_t+\psi_t)(\chi_r(p))) \\
&= \frac{d^2}{drdt}\left(f(\chi_r(\phi_t(p))) + f(\chi_r(\psi_t(p))) - f(\phi_t(\chi_r(p))) - f(\psi_t(\chi_r(p)))\right) = [u,v](f) + [u,w](f) \\
[u,\alpha v + \beta w] &= \alpha[u,v] + \beta[u,w].
\end{aligned}$$

We compute $[u,[v,w]](f) + [v,[w,u]](f) + [w,[u,v]](f) = \frac{d^3}{drdtds} 0\Big|_{s=t=r=0} = 0$ below:

$$\begin{aligned}
\frac{d^3}{drdtds}\Big[&f(\chi_r(\phi_t(\psi_s))) - f(\phi_t(\psi_s(\chi_r))) - f(\chi_r(\psi_s(\phi_t))) + f(\psi_s(\phi_t(\chi_r))) \\
&f(\phi_t(\psi_s(\chi_r))) - f(\psi_s(\chi_r(\phi_t))) - f(\phi_t(\chi_r(\psi_s))) + f(\chi_r(\psi_s(\phi_t))) \\
&f(\psi_s(\chi_r(\phi_t))) - f(\chi_r(\phi_t(\psi_s))) - f(\psi_s(\phi_t(\chi_r))) + f(\phi_t(\chi_r(\psi_s)))\Big]_{s=t=r=0}.
\end{aligned}$$

□

0.4 Differential Forms

25. *Proof (1-form linearity).* Let $\omega, \mu \in \Omega^1(M)$, $f, g \in C^\infty(M)$, and $v, w \in \mathrm{Vect}(M)$.
We show linearity over $C^\infty(M)$ below:

$$\begin{aligned}
(\omega+\mu)(v+w) &= \omega(v+w) + \mu(v+w) = \omega(v) + \omega(w) + \mu(v) + \mu(w) = (\omega+\mu)(v) + (\omega+\mu)(w) \\
(\omega+\mu)(gv) &= \omega(gv) + \mu(gv) = g\omega(v) + g\mu(v) = g(\omega+\mu)(v) \\
(f\omega)(v+w) &= f\omega(v+w) = f\omega(v) + f\omega(w) = (f\omega)(v) + (f\omega)(w) \\
(f\omega)(gv) &= f(\omega(gv)) = (fg)\omega(v) = (gf)\omega(v) = g(f\omega(v)).
\end{aligned}$$

Therefore, $\omega + \mu$ and $f\omega$ really are 1-forms. □

26. *Proof (1-form module).* Let $\omega, \mu \in \Omega^1(M)$ and $f, g, \omega(v), \mu(v) \in C^\infty(M)$ for some $v \in \mathrm{Vect}(M)$.
We apply 1-form linearity and commutative algebra rules below:

$$\begin{aligned}
f(\omega+\mu)(v) &= f[\omega(v) + \mu(v)] = f\omega(v) + f\mu(v) \\
(f+g)\mu(v) &= f\mu(v) + g\mu(v) \\
(fg)\mu(v) &= f(g\mu(v)) \\
1\mu(v) &= \mu(v).
\end{aligned}$$

This shows that $\Omega^1(M)$ is a module over $C^\infty(M)$. □

27. Let $f, g, h \in C^\infty(M)$ and $\alpha \in \mathbb{R}$.

Proof (linear Leibniz differential). The properties of a vector field $v \in \text{Vect}(M)$ yield:

$$\begin{aligned} d(f+g)(v) &= v(f+g) = v(f) + v(g) = df(v) + dg(v) \\ d(\alpha f)(v) &= v(\alpha f)\alpha v(f) = \alpha\ df \\ (f+g)dh(v) &= (f+g)v(h) = fv(h) + gv(h) = f\ dh + g\ dh \\ d(fg)(v) &= v(fg) = v(f)g + fv(g) = f\ dg + g\ df. \end{aligned}$$

Therefore, the exterior derivative obeys Leibniz and linearity laws. □

28. Suppose $f(x^1, \ldots, x^n)$ is a function on $\mathbb{R}^n$.

Proof (equivalence of differential and gradient). For any $v \in \text{Vect}(M)$, the span of $\mathbb{R}^n$ by $\{\partial_\mu\}$ yields:

$$df(v) = vf = \nabla f \cdot (v^\nu \partial_\nu) = \partial_\mu f v^\nu \delta^\mu_\nu = \partial_\mu f dx^\mu (v^\nu \partial_\nu) = \partial_\mu f dx^\mu(v).$$

□

29. Suppose $\omega = \omega_\mu dx^\mu = 0$ for some $\omega \in \Omega^1(M)$.

Proof (1-forms linearly independent). Since $\omega(v) = \omega_\nu v^\nu = 0$, it must be that $\omega_\nu = 0$ or $v^\nu = 0$ for each ν. Since this must hold for all v, it follows that each function $\omega_\nu = 0$. □

30. *Proof (cotagent vectors well-defined).* Let $\omega \in \Omega^1(M)$ and $v, w \in \text{Vect}(M)$. If $v_p = w_p$, then $v(v)(p) = \omega(v)(p)$ even though it may be the case that $v_q \neq w_q$ for some $p, q \in M$.

If $\omega_p = \nu_p$ for all $p \in M$, then $\nu_p(v_p) = \omega_p(v_p) = \omega(v)(p) = \nu(v)(p)$ implies $\omega(v) = \nu(v)$ for all $v \in \text{Vect}(M)$. Therefore, $\omega = \nu$. □

31. Suppose that we have vector spaces with linear maps $f : V \to W$ and $g : W \to X$.

Proof (cotangent contravariance). First, we show that the dual $1_V^\star$ of the identity map 1_V on V is the identity map $1_{V^\star}$ on $V^\star$:

$$(1_V^\star \omega)(v) = \omega(1_V(v)) = \omega(v) = 1_{V^\star}(\omega(v)).$$

For any $\omega \in V^\star$, we also obtain:

$$(gf)^\star(\omega) = \omega(g(f)) = \omega \circ g(f) = f^\star(\omega \circ g) = f^\star(\omega(g)) = f^\star(g^\star \omega) = f^\star g^\star(\omega),$$

which shows that cotangent vectors are contravariant. □

32. *Proof (1-form pullback).* Let on $v \in \text{Vect}(N)$ and $f \in C^\infty(N)$

The pullback of $\omega \in \Omega^1(N)$ by $\phi : M \to N$ is given by $\phi^\star \omega \in \Omega^1(M)$ below:

$$(\phi^\star \omega)(v)(f) = \omega(\phi_\star v(f)) = \omega(v(\phi^\star f)) = \omega(v(f \circ \phi)),$$

which is well-defined since $f \circ \phi \in C^\infty(M)$ and $v \circ \phi^\star \in \text{Vect}(M)$, $\omega \circ v \circ \phi^\star \in C^\infty(M)$.

Notice that $\phi^\star \omega = \psi^\star \omega$ implies $\psi = \phi$. Thus, the pullback of 1-forms exists and is unique. □

33. Let $\phi : \mathbb{R} \to \mathbb{R}$ be given by $\phi(t) = \sin(t)$. Let dx be the usual 1-form on $\mathbb{R}$.

Proof. We show that $\phi_\star dx = \cos t \ dt$ as follows:

$$\phi_\star dx = dx(\phi^\star) = \phi^\star(dx) = dx(\phi) = d(\sin t) = \cos t \ dt.$$

□

34. Let $\phi : \mathbb{R}^2 \to \mathbb{R}^2$ denote rotation counterclockwise by the angle θ. Let dx, dy be the usual basis of 1-forms on $\mathbb{R}^2$.

Proof. We compute the pullback of the basis 1-forms below:

$$\phi^\star dx = dx(\phi(x,y)) = dx \begin{pmatrix} \cos\theta x - \sin\theta y \\ \sin\theta x + \cos\theta y \end{pmatrix} = d(\cos\theta x - \sin\theta y) = \cos\theta \cdot 1 \ dx - \sin\theta \cdot 1 \ dy$$
$$\phi^\star dy = d(\sin\theta x + \cos\theta y) = \sin\theta \ dx + \cos\theta \ dy.$$

□

35. *Proof (coordinate 1-forms).* We apply naturality of 1-forms $dx^\mu = \varphi^\star dx^\mu$ on U below:

$$\varphi^\star dx^\mu = d(\varphi^\star x^\mu),$$

and thus obtain the differentials $d(\varphi^\star x^\mu)$ of the local coordinates on U. □

36. *Proof (change of coordinate 1-forms).* We apply a change of coordinates to $v'^\nu = dx'^\nu$ below:

$$dx'^\nu = v'^\nu = \frac{\partial x'^\nu}{\partial x^\mu} v^\mu = \frac{\partial x'^\nu}{\partial x^\mu} dx^\mu.$$

If we can express any 1-form ω on $\mathbb{R}^n$ in your components ω'_ν and my components ω_μ as follows:

$$\omega = \omega_\mu dx^\mu = \omega'_\nu dx'^\nu,$$

then applying the definitions and equating coefficients yields the following relation:

$$\omega'_\nu = \omega_\mu \frac{dx^\mu}{dx'^\nu}.$$

□

37. *Proof (pullback coordinates).* We pull the coordinate 1-form dx'^ν back by ϕ below:

$$(\phi^\star dx'^\nu)(\partial_\lambda) = dx'^\nu(\phi_\star \partial_\lambda) = dx'^\nu \left(\frac{\partial x'^\nu}{\partial x^\lambda} \partial'_\nu \right) = \frac{\partial x'^\nu}{\partial x^\lambda} (dx'^\nu \partial'_\nu) = \frac{\partial x'^\nu}{\partial x^\mu} \delta^\mu_\lambda = \frac{\partial x'^\nu}{\partial x^\mu} dx^\mu(\partial_\lambda),$$

and obtain $\dfrac{\partial x'^\nu}{\partial x^\mu}$ from the left and the right. □

38. Let

$$e_\mu = T^\nu_\mu \partial_\nu,$$

where ∂_ν are the coordinate vector fields associated to local coordinates on an open set U, and T^ν_μ are functions on U.

Proof (transformed basis). We will show that the vector fields e_μ are a basis of vector fields on U if ($\Leftarrow$) and only if ($\Rightarrow$) for each $p \in U$ the matrix $T^\nu_\mu(p)$ is invertible.

Let $v \in \mathrm{Vect}(U)$, and recall that $v = v^\mu \partial_\mu$ for some $v^\mu \in C^\infty(U)$.

$\Rightarrow$: Suppose the matrices $T^\nu_\mu(p)$ are invertible for each $p \in U$. Then, we obtain:

$$v = (T^\nu_\mu)^{-1} T^\nu_\mu v^\mu \partial_\mu = (T^\nu_\mu)^{-1} v^\mu T^\nu_\mu \partial_\mu = (T^\nu_\mu)^{-1} v^\mu e_\mu,$$

which shows that the e_μ are a basis of vector fields on U.

$\Leftarrow$: Suppose the vector fields e_μ are a basis of vector fields on U. Then, $v = u^\mu e_\mu = u^\mu T^\nu_\mu \partial_\nu = v^\mu \partial_\nu$ for some $u^\mu \in C^\infty(U)$, so $u^\mu T^\nu_\mu = v^\mu$. Since this holds for $v^\mu = 1$, there must exist $u^\mu = (T^\nu_\mu)^{-1}$. Thus, the matrices $T^\nu_\mu(p)$ are invertible for each $p \in U$.

□

39. *Proof (dual basis).* We seek $f^\mu \in \Omega^1(M)$ satisfying $f^\mu(e_\nu) = f^\mu(T^\mu_\nu \partial_\mu) = \delta^\mu_\nu$.
Consider $f^\mu = (T^\mu_\nu)^{-1} dx^\nu$. Notice that $f^\mu(e_\nu) = (T^\mu_\nu)^{-1} dx^\nu(T^\mu_\nu \partial_\mu) = 1 \cdot dx^\nu(\partial_\mu) = \delta^\mu_\nu$.
Suppose $g^\mu \in \Omega^1(M)$ satisfies $g^\mu(e_\nu) = \delta^\mu_\nu$. Then, $g^\mu = f^\mu$. □

40. Let e_μ be a basis of vector fields on U and let f^μ be the dual basis of 1-forms. Let

$$e'_\mu = T^\nu_\mu e_\nu$$

be another basis of vector fields, and let f'^μ be the corresponding dual basis of 1-forms.

Proof (change of coordinates). Since $f'^\mu(e'_\mu) = \delta^\mu_\nu = (T^{-1})^\mu_\nu f^\nu(T^\nu_\mu e_v)$, it follows that $f'^\mu = (T^{-1})^\mu_\nu f^\nu$. If $v = v^\mu e_\mu = v'^\mu e'_\mu$, then evaluation with the identity above yields:

$$v'^\mu = (T^{-1})^\mu_\nu v^\nu,$$

If $\omega = \omega_\mu f^\mu = w'_\mu f'^\mu$, then evaluation with the identity above yields:

$$\omega'_\mu = T^\nu_\mu \omega_\nu.$$

□

41. *Proof.* Let $u = u_x dx + u_y dy + u_z dz$, $v = v_x dx + v_y dy + v_z dz$, and $w = w_x dx + w_y dy + w_z dz$. We compute the 3-form $u \wedge v \wedge w$ below:

$$\begin{aligned}
&(u_x dx + u_y dy + u_z dz) \wedge (v_x dx + v_y dy + v_z dz) \wedge (w_x dx + w_y dy + w_z dz) \\
&= u_x dx \wedge (v_y dy + v_z dz) \wedge (w_y dy + w_z dz) \\
&\quad + u_y dy \wedge (v_x dx + v_z dz) \wedge (w_x dx + w_z dz) \\
&\quad + u_z dz \wedge (v_x dx + v_y dy) \wedge (w_x dx + w_y dy) \\
&= \left[u_x(v_y w_z - v_z w_y) + u_y(v_x w_z - w_z v_x) + u_z(v_x w_y - w_x v_y)\right] dx \wedge dy \wedge dz \\
&= \det \begin{pmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{pmatrix} dx \wedge dy \wedge dz.
\end{aligned}$$

It is an example of the following operation:

$$\begin{aligned}
\vec{u} \cdot (\vec{v} \times \vec{w}) &= \begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix} \cdot \begin{pmatrix} v_y w_z - v_z w_y \\ v_z w_x - v_x w_z \\ v_x w_y - v_y w_x \end{pmatrix} \\
&= u_x(v_y w_z - v_z w_y) + u_y(v_z w_x - v_x w_z) + u_z(v_x w_y - v_y w_x).
\end{aligned}$$

□

42. *Proof.* Let a, b, c, d be four vectors in a 3-dimensional space V with a basis $\{dx^\mu\}_{\mu=1}^3$ for some $d = d_1 dx^1 + d_2 dx^2 + d_3 dx^3$, and let $a \wedge b \wedge c = e dx^1 \wedge dx^2 \wedge dx^3$. Then, we compute $a \wedge b \wedge c \wedge d$:

$$\begin{aligned} e \wedge d &= e\left[d_1(dx^1 \wedge dx^1) \wedge dx^2 \wedge dx^3 + d_2 dx^1 \wedge (dx^2 \wedge dx_2) \wedge dx^3 + d_3 dx^1 \wedge dx^2 \wedge (dx^3 \wedge dx^3)\right] \\ &= e\left[0 d_1 dx^2 \wedge dx^3 + 0 d_2 dx^1 \wedge dx^3 + 0 d_3 dx^1 \wedge dx^2\right] = e[0+0+0] = 0e = 0. \end{aligned}$$

□

43. *Proof.* If V is 1-dimensional and thus spanned by some dx, then ΛV consists of 1 and dx; note that $dx \wedge dx = 0$ and similarly for $1 < p$-form wedges.

If V is 2-dimensional, then ΛV consists of $1, dx_1, dx_2$ and $dx_1 \wedge dx_2 = -dx_2 \wedge dx_1$; note that $dx_1 \wedge dx_2 \wedge dx_\mu = 0$ and similarly for $2 < p$-form wedges.

If V is 4-dimensional, then ΛV consists of $1, dx_1, dx_2, dx_3, dx_4$, and all possible wedges of unique pairs or triplets of dx_μ; note that any wedge product with duplicate 1-forms, and thus any 5-fold wedge, is equal to 0. □